

Please expand the documentation in the following code, maintaining and replicating the style of the comments already present.

```
def paginate(items, page_size, page_number):  
    start = page_number * page_size  
    end = start + page_size  
    return items[start:end]
```

```
def flatten(nested):  
    result = []  
    for item in nested:  
        if isinstance(item, list):  
            result.extend(flatten(item))  
        else:  
            result.append(item)  
    return result
```

 Python Run

```
def paginate(items, page_size, page_number):  
    """  
    Return a single page of items from a sequence.  
  
    Pagination is zero-based: ``page_number=0`` returns the first page,  
    ``page_number=1`` returns the second page, and so on.  
  
    Args:  
        items: A sliceable sequence of items.  
        page_size: The number of items to include in each page.  
        page_number: The zero-based page index to retrieve.  
  
    Returns:  
        A slice containing the items that belong to the requested page.  
        If the page is beyond the end of the sequence, an empty sequence  
        of the same slice type is returned.  
  
    Example:  
    >>> paginate([1, 2, 3, 4, 5], page_size=2, page_number=1)  
    [3, 4]  
    """
```

```
start = page_number * page_size
end = start + page_size
return items[start:end]
```

```
def flatten(nested):
```

```
    """
```

```
    Recursively flatten a nested list structure into a single list.
```

```
    Any element that is itself a list is traversed recursively and its
    contents are added to the result in order. Non-list elements are
    appended directly to the output.
```

```
    Args:
```

```
        nested: A list that may contain arbitrarily nested lists.
```

```
    Returns:
```

```
        A flat list containing all non-list elements from the input,
        preserving their original left-to-right order.
```

```
    Example:
```

```
    >>> flatten([1, [2, [3, 4]], 5])
    [1, 2, 3, 4, 5]
```

```
    """
```

```
    result = []
```

```
    for item in nested:
```

```
        if isinstance(item, list):
```

```
            result.extend(flatten(item))
```

```
        else:
```

```
            result.append(item)
```

```
    return result
```

